

Pulkit Garg

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EDUCATION

University of Waterloo - Bachelor of Computer Science

Sep 2022 - Present

Cognitive Science Minor · Fine Arts Specialization · President's Scholarship of Distinction

TECHNICAL SKILLS

Languages: C++, C#, Python, Java, Kotlin, GLSL, HLSL

Graphics & Systems: Vulkan, OpenGL, GPU/Compute Programming, Multithreading, Performance Optimization, Real-Time Rendering

Tools & Frameworks: Git, CMake, Android SDK, Unity, Unreal Engine, React Native, Firebase

EXPERIENCE

Graphics Programming Intern - SideFX

Jan 2026 - Present

- Developed core, performance-critical, cross-platform Vulkan viewport functionality for Houdini, creating reusable rendering libraries and custom GLSL shader infrastructure for professional VFX workflows.
- Optimized instanced-scene rendering to achieve more than 20x polygon reduction across instances while substantially reducing draw calls for complex viewport scenes.
- Designed and built a GPU-based Vulkan text-rendering pipeline from scratch using bitmap fonts and texture atlases; implemented skyline bin-packing and extended the pipeline for TrueType font rasterization.

3D Software Development R&D Intern - SideFX

May 2025 - Sep 2025

- Strengthened scientific and simulation data-import workflows by implementing high-precision numerical parsing and processing.
- Reworked mesh-decimation algorithms to achieve more than 10x polygon reduction while preserving topology and associated UV/texture maps.
- Designed and implemented a real-time Push-Pull geometry-editing tool with custom spatial acceleration structures, reducing user-interaction complexity by up to 12x and increasing modeling throughput by 30%.

Virtual Reality Developer - Liftwerx

Sep 2024 - Dec 2024

- Delivered VR training modules for wind-turbine technicians using Unity, Unreal Engine, and Blender.
- Owned end-to-end scripting, debugging, and deployment to Meta Quest headsets through a custom Android launcher and AWS S3 distribution pipeline.
- Engineered a custom Android plugin for centralized data synchronization and real-time technician-performance reporting.

Software Developer Intern - OS & Graphics - Ford Motor Company of Canada

Jan 2024 - Apr 2024

- Built C++ and Kotlin integrations for lighting and climate vehicle systems within Android Automotive OS.
- Introduced multithreaded service execution, reducing service load times by up to 70%; also reduced memory usage and addressed memory leaks.

Software Developer Intern - 3D Visualization - Ford Motor Company of Canada

May 2023 - Sep 2023

- Developed real-time 3D visualization tools for next-generation Ford vehicles with Google Filament; implemented asynchronous asset loading to improve load time, graphical fidelity, and user experience.

PROJECTS

FluidSim - C++, Vulkan, GLSL

GitHub

Built a Vulkan application that simulates water with smoothed particle hydrodynamics (SPH); fluid dynamics, neighbor search, and whitewater in GPU compute shaders. Implemented a spatial hash grid for neighbor queries and screen-space surface reconstruction with reflection, refraction, depth absorption, and shadows.

Voksel - C#, HLSL

GitHub

Built a cross-platform, open-source 3D modeling tool for rapidly creating low-poly game and level-design assets. Implemented import/save, FBX/OBJ export, configurable color palettes, and Unity/Blender control presets.

SpaceLine - Unity, SteamVR

GitHub

Built a physics-based space sandbox with real-time gravity and climate simulation; added a SteamVR mode for immersive exploration. Won NASA Space Apps India and was selected as a global finalist.

Prepto - React Native, Firebase

GitHub

Built a cross-platform app for real-time competitive test preparation with Firebase authentication and matchmaking. Synced users, questions, games, and time-stamped responses in Firebase to support real-time game state.